(3 Hours) [Total Marks: 80]

Instructions:

N.B. : (1) Question No 1 is Compulsory.
(2) Attempt any three questions out of the remaining five.
(3) All questions carry equal marks.
(4) Assume suitable data, if required and state it clearly.

Q1. Attempt any 4 questions out of 6. Each question carries 5 marks.

a) Compare between DOS, NOS and Middleware.
b) Explain with the diagram Dispatcher-Worker thread model.
c) Explain happens before relation with its features.
d) Explain naming in a distributed system.
e) Compare caching and replication.
f) Explain stream - oriented communication.

Q2. Each question carries 10 marks.

i. Explain in detail Raymond's Tree-Based algorithm (Token-based algorithm).
ii. Write a note on the Network File System (NFS) .

Q3. Each question carries 10 marks.

i. Write note on Andrew File System (AFS) .
ii. Explain steps involved in the RMI execution process in detail.

Q4. Each question carries 10 marks.

i. What is Remote Procedure Call?Explain the working of RPC in detail.
ii. Explain different types of distributed systems with diagrams.

Q5. Each question carries 10 marks.

i. Explain with diagrams various client-centric consistency models.
ii. Compare static and dynamic load balancing algorithms.

Q6. Each question carries 10 marks.

i. Explain with an example load sharing approach.
ii. Explain any one election algorithm in detail with suitable example.